



SMARTSPLIT

Intelligent Bill Splitting & Expense Sharing Application

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Problem Statement

Traditional bill-splitting methods are inefficient and prone to errors. Current apps lack real-time automation, AI features, and seamless settlement systems, creating confusion and disputes among users.

Introduction

SmartSplit is a cross-platform mobile application designed to streamline shared expense management among friends, flatmates, couples, and travelers. By leveraging modern AI and real-time technology, it eliminates manual calculation errors and provides intelligent automation for seamless bill splitting.

Tech Stack

Flutter	Firebase
Node.js	Tesseract OCR
Google Vision	Android SDK

System Workflow



Key Features

- AI-powered receipt scanning with OCR technology
- Multi-user group expense tracking
- Optimized settlement algorithm (graph-based)
- Visual spending insights & analytics
- Real-time notifications for dues & expenses
- End-to-end encryption for financial data
- Cross-platform synchronization
- Spending pattern analysis

Project Timeline (Gantt Chart)

M1: Proposal Poster (10%)	Nov 17
Planning & Research	Nov-Dec
System Design & Prototyping	Dec-Jan
M2: Mid-Project Review (20%)	Jan 12-25
Core Development	Jan-Mar
AI/OCR Integration	Mar-Apr
Testing & Refinement	Apr-May
Report Writing	Apr-May
M3: Artefact Demo (80%)	May
M4: Final Submission	May

AI Integration

OCR Technology: Automated receipt scanning extracts merchant name, date, items, prices, and totals using Tesseract and Google Vision APIs.

Smart Features: Predictive expense categorization and spending pattern analysis powered by machine learning algorithms.

Aims & Objectives

- Secure authentication system
- Group & member management
- AI-powered OCR receipt scanning
- Automatic balance calculation
- Real-time synchronization
- Visual spending analytics
- Data encryption & security
- Modern responsive UI/UX

Expected Outcomes

- Fully functional MVP
- Real-time expense sharing
- OCR-enhanced scanning
- Group synchronization
- Live analytics dashboard
- Polished UI/UX

Challenges

- OCR accuracy across receipt formats
- Real-time synchronization
- Secure financial data storage
- Concurrent group modifications